

RAPID CITY, SD, USA

WMO: 726620

Lat: 44.046N

Lon: 103.054W

Elev: 3160

StdP: 13.09

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 24090

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -8.4	(c) -3.0	(d) -14.9	(e) 2.8	(f) -7.9	(g) -9.4	(h) 3.7	(i) -2.1	(j) 41.4	(k) 28.6	(l) 37.7	(m) 28.0	(n) 7.4	(o) 350	(p) 0.669

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 26.8	(c) 96.6	(d) 66.1	(e) 92.4	(f) 65.9	(g) 88.8	(h) 65.3	(i) 71.3	(j) 85.3	(k) 69.6	(l) 84.3	(m) 68.0	(n) 83.1	(o) 12.7	(p) 170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 67.1	(b) 112.3	(c) 78.6	(d) 65.1	(e) 104.4	(f) 76.4	(g) 63.2	(h) 97.6	(i) 74.5	(j) 37.2	(k) 85.4	(l) 35.6	(m) 84.3	(n) 34.1	(o) 82.7	(p) 78.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
35.4	30.6	26.1	-16.6	102.2	5.3	4.4	-20.5	105.4	-23.6	108.0	-26.6	110.5	-30.5	113.7
			-16.8	74.5	4.7	2.2	-20.2	76.1	-22.9	77.4	-25.5	78.7	-28.9	80.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.3	24.9	26.4	35.5	44.4	54.4	65.1	73.3	71.5	62.0	47.9	35.4	25.7
	DBStd	20.14	13.32	14.04	12.98	10.03	8.49	7.49	6.30	6.17	9.03	9.89	11.74	12.79
	HDD50	3553	778	664	466	219	53	2	0	0	13	155	449	754
	HDD65	7113	1243	1082	915	619	342	87	8	17	160	534	888	1218
	CDD50	2575	0	2	16	51	188	455	724	666	372	90	11	0
	CDD65	660	0	0	0	1	12	90	267	218	69	3	0	0
	CDH74	8551	0	0	16	71	239	1189	3212	2577	1106	136	5	0
	CDH80	3870	0	0	1	12	60	467	1601	1222	470	37	0	0
(14) Wind	WSAvg	10.3	10.1	10.8	11.3	12.3	11.4	9.9	9.1	9.0	9.7	10.7	10.0	9.7
(15) Precipitation	PrecAvg	17.7	0.4	0.4	0.9	2.2	3.5	2.9	2.2	1.9	1.3	1.5	0.5	0.3
	PrecMax	23.7	1.0	0.9	1.9	4.8	7.3	6.3	3.9	4.2	3.0	5.4	1.3	1.1
	PrecMin	12.2	0.0	0.0	0.0	0.4	1.5	0.7	0.6	0.8	0.2	0.1	0.0	0.0
	PrecStd	3.5	0.3	0.3	0.5	1.0	1.7	1.6	0.9	0.9	0.7	1.3	0.4	0.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.6	67.1	76.2	81.6	86.5	97.8	103.5	99.7	96.2	84.6	73.2	61.2
		MCWB	43.5	47.2	50.9	55.0	62.9	65.1	66.9	66.1	63.8	57.6	49.9	44.9
	2%	DB	53.6	58.2	68.3	74.8	80.5	90.3	97.5	95.4	90.7	77.5	64.8	53.3
		MCWB	40.7	42.9	48.4	53.6	60.9	66.3	67.1	65.5	62.3	55.3	46.5	40.6
	5%	DB	48.1	51.6	62.1	68.4	75.4	85.1	93.0	91.1	85.5	71.0	59.1	47.9
		MCWB	38.0	39.9	46.0	50.8	59.0	65.8	67.2	65.5	61.3	52.3	44.4	37.4
	10%	DB	43.3	45.5	55.7	62.6	71.1	80.7	88.9	87.1	80.6	64.9	53.0	42.7
		MCWB	35.3	36.3	42.9	48.2	56.8	64.0	66.4	65.4	59.5	50.0	41.5	34.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.8	47.6	52.4	57.5	66.4	73.1	74.5	72.8	68.5	60.0	50.8	45.4
		MCDB	59.0	64.8	71.8	75.2	80.5	84.3	87.9	86.4	86.0	77.9	71.1	60.0
	2%	WB	41.3	43.6	49.3	54.4	62.9	69.9	72.0	70.2	65.6	56.5	47.5	41.1
		MCDB	52.6	57.2	66.7	71.7	76.3	82.5	86.2	85.0	82.8	73.5	63.3	52.5
	5%	WB	38.5	40.2	46.5	51.8	60.2	67.8	70.3	68.5	63.3	53.7	44.6	38.0
		MCDB	47.3	50.5	61.1	66.8	73.1	81.6	85.1	83.7	80.9	68.0	58.0	47.1
	10%	WB	35.7	36.9	43.3	49.1	57.6	65.6	68.5	66.9	61.0	50.9	41.8	35.1
		MCDB	42.9	44.9	55.1	61.2	68.7	78.8	83.8	81.9	77.3	63.5	52.5	42.4
(35) Mean Daily Temperature Range	MDBR		22.0	22.6	24.1	23.8	23.2	24.7	26.8	27.5	27.7	25.0	23.8	21.5
	5% DB	MCDBR	28.9	31.5	35.4	35.6	31.0	32.3	33.3	33.8	35.6	35.7	33.3	28.1
		MCWBR	18.4	18.8	19.0	18.4	15.8	14.4	11.9	12.5	14.4	17.5	18.5	17.6
	5% WB	MCDBR	27.7	29.8	33.5	32.5	28.8	28.8	27.5	28.2	31.6	32.1	31.5	27.2
		MCWBR	18.2	18.2	18.5	17.4	15.5	15.1	12.5	12.3	13.6	16.2	17.9	17.5
(40) Clear-Sky Solar Irradiance	taub		0.253	0.262	0.280	0.315	0.333	0.346	0.369	0.364	0.327	0.290	0.263	0.252
	taud		2.577	2.551	2.548	2.436	2.462	2.451	2.419	2.415	2.499	2.577	2.564	2.563
	Ebn at Noon		287	302	308	301	297	292	284	282	286	287	280	277
	Edh at Noon		20	24	28	34	34	35	36	35	29	23	20	18
(44) All-Sky Solar Radiation	RadAvg		611	916	1286	1596	1893	2200	2234	1922	1521	1000	676	510
	RadStd		32	53	100	94	181	161	82	85	94	95	50	38

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page